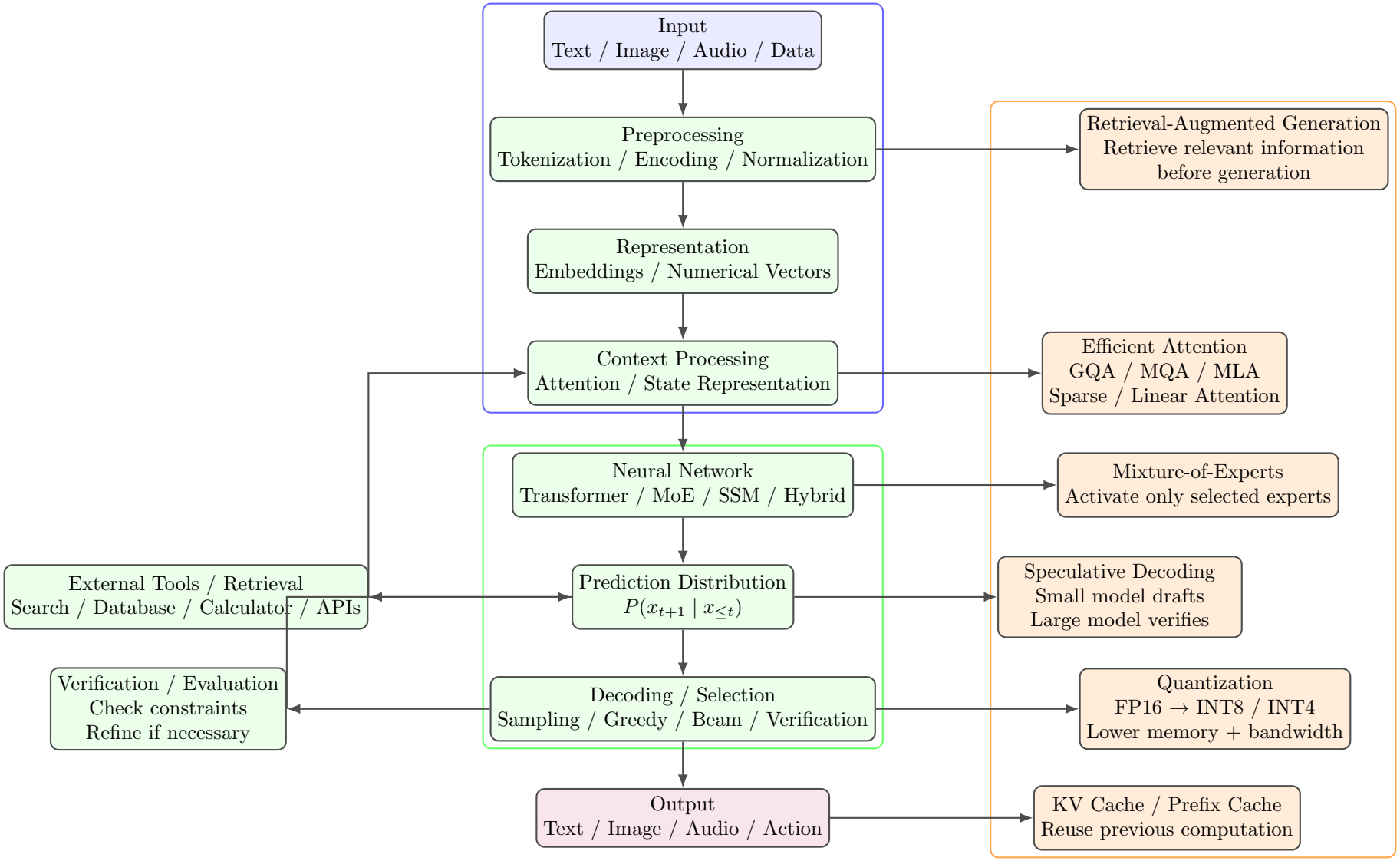

Input
Text / Image / Audio / Data
Preprocessing
Tokenization / Encoding / Normalization
Representation
Embeddings / Numerical Vectors
Context Processing
Attention / State Representation
Neural Network
Transformer / MoE / SSM / Hybrid
Prediction Distribution
$P(x_{t+1} \mid x_{\leq t})$
Decoding / Selection
Sampling / Greedy / Beam / Verification
Output
Text / Image / Audio / Action
External Tools / Retrieval
Search / Database / Calculator / APIs
Verification / Evaluation
Check constraints
Refine if necessary
Retrieval-Augmented Generation
Retrieve relevant information
before generation
Efficient Attention
GQA / MQA / MLA
Sparse / Linear Attention
Mixture-of-Experts
Activate only selected experts
Speculative Decoding
Small model drafts
Large model verifies
Quantization
FP16 → INT8 / INT4
Lower memory + bandwidth
KV Cache / Prefix Cache
Reuse previous computation